HW 03 - Road traffic accidents

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Figure 1: Photo by Clark Van Der Beken on Unsplash

In this assignment we'll look at traffic accidents in Edinburgh. The data are made available online by the UK Government. It covers all recorded accidents in Edinburgh in 2018 and some of the variables were modified for the purposes of this assignment.

Getting started

Learning Objectives

In this assignment, you will practice:

- Creating density plots to show distributions over time
- Using `mutate()` and `if_else()` to create new categorical variables
- Faceting plots to compare groups
- Interpreting visualizations in context of data
- Designing your own multi-variable visualization

Navigate to your `homework-instructions` repo in JupyterHub, open the `HW03` folder, and open the R Markdown document `hw-03.Rmd`. Knit it to make sure it compiles without errors.

Warm up

Before we introduce the data, let's warm up with some simple exercises.

Complete these steps:

1. Update the YAML, changing the author name to your name
2. **Knit** the document
3. **Commit** your changes with message: `"Updated author name"`
4. **Push** to GitHub
5. Verify your changes are visible in your GitHub repo (check both .Rmd and .md files)

If anything is missing, commit and push again.

Packages

We'll use the **tidyverse** package for much of the data wrangling and visualization and the data lives in the **dsbox** package. These packages are already installed for you. You can load them by running the following in your Console:

```r
library(tidyverse)
library(dsbox)
```

Data

The data can be found in the **dsbox** package, and it's called `accidents`. Since the dataset is distributed with the package, we don't need to load it separately; it becomes available to us when we load the package. You can find out more about the dataset by inspecting its documentation, which you can access by running `?accidents` in the Console or using the Help menu in RStudio to search for `accidents`. You can also find this information here.

Exercises

1. How many observations (rows) does the dataset have? Instead of hard coding the number in your answer, use inline code.

   There are 768 observations in the dataset

2. Run `View(accidents)` in your Console to view the data in the data viewer. What does each row in the dataset represent?

Each row in the dataset represents a set of information about a distinct accident. Some variables in each row include longitude, latitude, severity, and casualities.

Knit → Commit with message `"Completed Exercises 1-2"` → Push

3. Create a plot that shows the distribution of accidents throughout the day, comparing weekdays vs. weekends and by accident severity.

   Your plot should include:

- A density plot showing `time` on the x-axis
- Different colors for different `severity` levels
- Separate facets for weekdays vs. weekends
- Appropriate labels and title

Step 1: Create a new variable called `day_of_week_type` that categorizes days as "Weekend" (Saturday and Sunday) or "Weekday" (all other days)

Hint: Use `mutate()` with `if_else()`. The condition should check if `day_of_week` is in the vector `c("Saturday", "Sunday")`

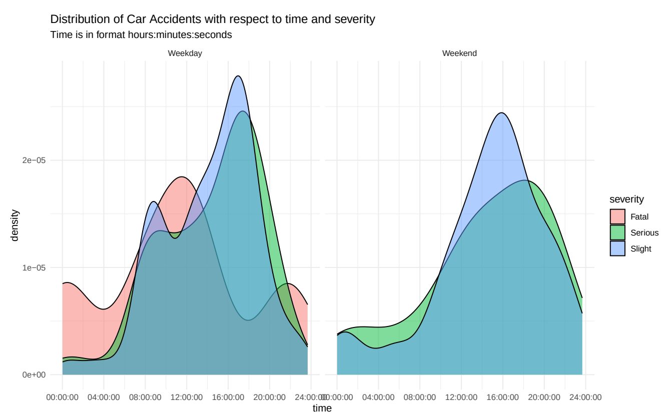
```r
accidents <- accidents %>%
  mutate(day_of_week_type = if_else(
```

```r
    day_of_week %in% c("Saturday", "Sunday"),
    "Weekend",
    "Weekday"
  ))
```

Step 2: Build the visualization using density plots
Hints:

- Use `geom_density()` with `alpha = 0.5` for transparency
- Use `fill = severity` in your aesthetic mapping to color by severity
- Use `facet_wrap()` to create separate plots for weekday vs. weekend
- Add `theme_minimal()` and appropriate labels

```r
ggplot(data = accidents, aes(x = time, fill = severity)) +
  geom_density(alpha = 0.5) +
  facet_wrap(~day_of_week_type, ncol = 2) +
  theme_minimal() +
  labs(
    title = "Distribution of Car Accidents with respect to time and severity",
    subtitle = "Time is in format hours:minutes:seconds",
    x = "time",
    y = "density",
    fill = "severity"
  )
```



Interpretation (2-3 sentences): Describe what patterns you observe:

- When do accidents tend to occur during weekdays vs. weekends?
- How does the severity of accidents vary by time of day?
- Are there any notable differences between weekdays and weekends?

During weekdays, accidents tend to occur between 8:00 am and 6:00 pm, whereas during the weekends, accidents tend to occur between noon and 6 pm. In other words, accidents tend to occur earlier (in the morning) during the weekdays. I notice that in this dataset, on the weekdays there are more fatal accidents than the weekends, which do not have any observations of fatal accidents.

After completing Exercise 3:

Knit → Commit with message `"Completed Exercise 3"` → Push

4. Create your own data visualization based on the accidents data. Your visualization must include at least three variables.

   Requirements:

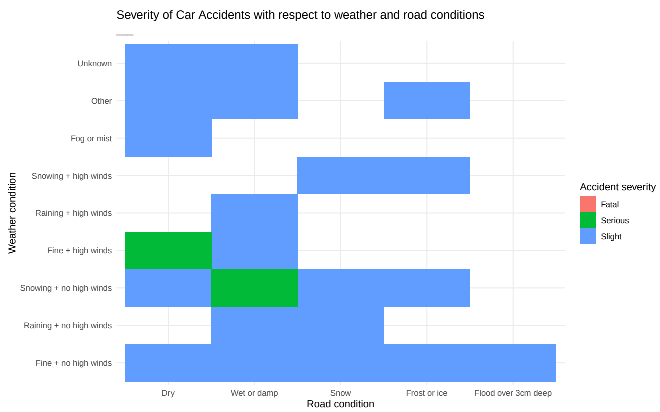
- Use at least 3 variables from the dataset
- Choose an appropriate plot type for your question
- Include proper labels (title, axis labels, legend if applicable)
- Write an interpretation of what the plot shows

  Some ideas to explore:

- How does accident severity vary by weather conditions and light conditions?
- What is the relationship between speed limit, number of vehicles, and severity?
- How do accident patterns differ across different road types and weather conditions?
- Do certain vehicle types have more accidents in specific conditions?

  Your visualization:

```r
ggplot(data = accidents, aes(x = weather, y = road_surface, fill = severity)) +
  geom_tile() +
  theme_minimal() +
  coord_flip() +
  labs(
    title = "Severity of Car Accidents with respect to weather and road conditions",
    subtitle = "___",
    x = "Weather condition",
    y = "Road condition",
    fill = "Accident severity"
  )
```



What question does your visualization answer?

How are different road and weather conditions associated with the severity of car accidents?

What variables were used:

I used road_surface, weather, and severity for my variables.

Interpretation (3-4 sentences): What patterns or insights does your visualization reveal?

I notice that most "serious" accidents occur when the road condition is either dry or wet or damp. I also notice that most "serious" accidents occur when the weather is either "Fine + high winds" or "Snowing + No high winds". Outside of these conditions it seems that most car accidents have slight severity.

After completing Exercise 4:

Knit → Commit with message `"Completed Exercise 4"` → Push

Common Errors and Troubleshooting

Data wrangling errors:

- **Error: "object 'day_of_week_type' not found"** → Make sure you ran the `mutate()` step and saved it back to `accidents` with `<-`
- **`if_else()` not working** → Check that you're using `%in%` for checking multiple values: `day_of_week %in% c("Saturday", "Sunday")`

Visualization errors:

- **Density plot looks wrong** → Make sure `time` is on the x-axis and `severity` is mapped to `fill`
- **Facets not appearing** → Check that you created the `day_of_week_type` variable first

- **Colors not showing** → Make sure `fill = severity` is inside `aes()`
- **Plot is too small/hard to read** → Try adjusting `fig.width` in your chunk options

General issues:

- **Error: "could not find function"** → Load the tidyverse package: `library(tidyverse)`
- **Changes not showing** → Save your file and knit again
- **Inline code not working** → Remember to use backticks around your R code

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To submit:

1. Click the **Knit** dropdown menu (next to the Knit button)
2. Select **Knit to tufte_handout** to generate a PDF
3. Download the PDF file from your Files pane
4. Upload the PDF to Canvas

Final check: Visit your GitHub repo to confirm all your commits are there!